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AMATEUR VOLLEYBALL TOURNAMENT

Description:

A sports center wants to build a simple system to organize and manage an amateur volleyball tournament. The goal is to model the basic elements of a tournament and handle operations related to teams, matches, and results.

Users:

- One Organizer
- Dozens Teams
- Hundreds Players
- Thousands Spectators
- Tens Referees
- Dozens Coaches

Requirements:

- R1 : The coaches can register the teams, with number and nominative of the players, and each team has its home stadium
- R2 : The tournament brackets, the teams participating and dates of the games are decided by the organizer
- R3 : The organizer must assign a referee for each match
- R4 : Spectators can view the dates of upcoming tournament's games
- R5 : Spectators can reserve a spot for each match
- R6 : Spectators can view the tournament scoreboard and check the results
- R7 : Each team can enter in more than one tournament
- R8 : Each match is assigned a stadium

Additional context:

- AC1 : Peaks of users are expected close to the dates of important matches
- AC2 : The organizer sets the tournament structure based on how many teams are registered or evolving needs of the organization

Important Architecture Characteristics:

- **Elasticity: Medium 3/5** - The system expects to have peaks of users near important matches, although never reaching a really significant number (thousands at most). As for requirement AC1.
- **Evolvability: High 4\5** - The organization or scope of an amatorial tournament could change in unexpected directions. The system should have the possibility to introduce updates like a subscription plan, planning tournaments for different sports or other functionalities, as indicated by AC2 and to a lesser extent by R2.
- **Modularity: Medium\High 3\5** - The system is expected to be divided into modules to manage different aspects of the system (like tournament organization, teams management…)
  As for the discrete components necessary for requirements R2, R4, R5, R6.
- **Overall cost: Low\Medium 2\5** - A system developed for an amatorial tournament between local teams wouldn't have significant budgets or years to be developed.
- **Responsiveness: Medium\High 4\5** - How quick is the system at doing the requested operations matters as there are real time interactions with users, like in requirements R4, R5 and R6.